Journal of Experimental Psychology:
Human Perception and Performance
2013, Vol. 39, No. 1, 133–155

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0096-1523/13/$12.00 DOI: 10.1037/a0028001

Variants of Independence in the Perception of Facial Identity and Expression

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A prominent theory in the face perception literature—the parallel-route hypothesis (Bruce & Young, 1986)—assumes a dedicated channel for the processing of identity that is separate and independent from the channel(s) in which nonidentity information is processed (e.g., expression, eye gaze). The current work subjected this assumption to experimental tests from the perspective of three characterizations of independence and separability: Garner's speeded classification task (Garner, 1974), general recognition theory (Ashby & Townsend, 1986), and systems factorial technology (Townsend & Nozawa, 1995). The goals of the study were to (a) provide converging evidence on validity of the parallel-route hypothesis, and (b) understand the relationships between various definitions of independence and separability. The results of the current work demonstrated that identity and expression are not completely independent, using data from all three perspectives. Consequently, a modification of the central assumption of the parallel-route hypothesis is required. In addition, the intricate but orderly relationships between various types of separability and independence have implications for theories of perceptual independence.

Keywords: face recognition, the parallel-route hypothesis, facial identity and expression, perceptual independence

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Faces are multidimensional visual stimuli, capable of transmitting a range of information, including identity, emotion, social intentions, and so forth. Accurate and rapid extraction of these multiple dimensions is crucial for efficient and effective social interaction. A major goal in psychology is to determine how these dimensions are processed and to what extent they might interact. A prominent theory in face perception—the parallel-route hypothesis (Bruce & Young, 1986; Burton, Bruce, & Johnston, 1990)—holds that identity is processed independently of other dimensions, such as expression. According to this account, the recognition of identity relies on invariant, long-term, abstract representations, stored in face-recognition units (FRUs), that are separate and independent of representations used to identify other aspects of the face.

The parallel-route hypothesis is based on a set of assumptions regarding both representational and functional aspects of face recognition (see Ingvalson & Wenger, 2005, for a related analysis). With respect to representation, the hypothesis assumes complete independence between the representation of identity and that of all other dimensions. With respect to function, the hypothesis assumes a parallel architecture with no interactions between the channels processing identity and any other dimension, such as expression. Although these assumptions have been adopted in other prominent models of face perception (Haxby, Hoffman, & Gobbini, 2000, 2002),[1] there are both theoretical arguments and empirical data that call these assumptions into question (see, e.g., Ganel & Goshen-Gottstein, 2002, 2004; Calder & Young, 2005). The goal of the present study is to provide a set of rigorous tests of these assumptions using three complementary metatheoretical approaches: Garner's logic of inferring informational structure (Garner, 1974), general recognition theory (GRT, Ashby & Townsend, 1986), and systems factorial technology (SFT, Townsend & Nozawa, 1995; Townsend & Wenger, 2004a, 2004b). The Garner approach and GRT address representational separability, GRT addresses representational independence, and SFT addresses assumptions regarding processing architecture and independence in rates of processing.

Evidence For and Against the Parallel-Route Hypothesis

Evidence in support of the assumptions of the parallel-route hypothesis comes from behavioral (Bruce & Young, 1986), neuropsychological (Hornak, Rolls, & Wade, 1996; Humphreys, Donnelly, & Riddoch, 1993; Kurcuz & Feldmar, 1979; Perrett, Rolls, & Caan, 1982; Perrett et al., 1984), computational (Calder, Burton, Miller, Young, & Akamatsu, 2001), and neuroimaging sources (George et al., 1999; Breiter, Etcoff, Kennedy, Rauch, & Buckner,

This article was published Online First May 7, 2012.

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We thank Robert Nosofsky, Daniel Algom, and two anonymous reviewers for their helpful comments on an earlier version of this paper. This work was completed while both authors were in residence at The Pennsylvania State University.

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[1] In fairness to Haxby et al. (2000), we note that their model claims for cross-talk between the two functions. However, in practice, and according to their own way of reckoning, the model is elusive with respect to the implementation and level of such cross-talk: "the degree of separation between the functional roles played by the different regions in this system is unclear" (p. 231).

1996). An extensive review of these lines of evidence can be found in Ganel and Goshen-Gottstein (2004) and in Calder and Young (2005). Here we present a brief summary of the most important evidence. The influential work by Ekman and Friesen (Ekman & Friesen, 1976) proposes that many facial expressions are perceived independently of specific cultural influences on the observer. Bruce (1986) has shown that prior personal familiarity with a face had no effects on reaction time (RT) in a task that called for expression judgments. Similarly, Etcoff (1984) found perfect selective attention (i.e., no interference) to either identity or expression in a speeded classification task.

Prosopagnosic patients have been described who are impaired in recognizing faces yet who are not impaired in their abilities to process expression (Humphreys et al., 1993; Parry et al., 1991); the complementary pattern has also been documented (Humphreys et al., 1993; Kurcuz & Feldmar, 1979). Neuroimaging studies have suggested that identity is associated with activity in the inferior occipital lateral fusiform gyri (George et al., 1999), whereas the processing of expression is primarily mediated by the amygdala (Breiter et al., 1996; Morris et al., 1996) and the superior temporal sulcus (STS, Desimone, 1991). The idea of independent processes mediating the perception of identity and expression is also supported by data from single-cell recording studies in monkey temporal cortex (Desimone, 1991; Hasselmo, Rolls, & Baylis, 1989).[2] These studies suggest that these facial dimensions are coded by separate cell populations (see also Sugase, Yamane, Ueno, & Kawano, 1999). A different tack was used by Calder et al. (2001). These researchers applied PCA (i.e., principal component analysis) to the faces used by Ekman and Friesen (1976). They concluded that identity and expression are represented by distinct statistical components. Finally, an independence in the processing of identity and expression is also suggested by behavioral data from healthy participants. For example, repetition-priming effects are found in tasks of familiarity judgments but not for expression judgments (Ellis, Young, & Flude, 1990).

There are, however, arguments and data that run counter to the hypothesis. For example, Calder and Young (2005) argued that many of the studies supporting independence of identity and expression suffer from a range of methodological flaws. These authors argue that the idea that prosopagnosics can recognize facial expressions is usually supported only by anecdotal reports; on formal testing, most such patients also show impairments of facial expression recognition (Calder & Young, 2005). As another example, Young (1996) describes an amygdalotomy patient, D.R., who was severely impaired in her ability to make expression judgments and to a lesser extent in her abilities to decide whether two pictures depicted same or different identities. However, D.R. showed severe performance deficits when presented with two pictures of the same identity with two different expressions. This pattern of performance suggests some cross-talk, in at least one direction, from identity to expression.

An important line of evidence against the parallel-route hypothesis, and one that provides the motivation for the current study, comes from application of Garner's speeded classification task (Garner, 1974). Schweinberger and colleagues (Kaufmann & Schweinberger, 2004; Schweinberger, Burton, & Kelly, 1999; Schweinberger & Soukup, 1998) applied the Garner speeded-classification task to the dimensions of identity and expression. These researchers found that the processing of identity interfered with the processing of expression (i.e., reliable levels of interference were obtained), while the reverse was not true. These results have been replicated in both healthy and schizophrenic populations (Baudouin, Martin, Tiberghien, Verlut, & Franck, 2002), with asymmetric pattern supporting an inference of partial integrality (Garner, 1974). Ganel and Goshen-Gottstein (2004) proposed that the absence of interference from expression to identity in previous studies is the result of an experimental artifact. Specifically, they argued that if the relative discriminability of the dimensions of identity and expression were equated, or expression was made more discriminable than identity, then expression should interfere with identity and vice versa. In addition, Ganel and Goshen-Gottstein correctly predicted that if found, this symmetric interference would be larger for familiar than for unfamiliar faces. All of this suggests channel cross-talk rather than independence.

Three Approaches to Characterizing Independence

The evidence reviewed above, in particular that coming from the Garner paradigm (Ganel & Goshen-Gottstein, 2002, 2004; Kaufmann & Schweinberger, 2004; LeGal & Bruce, 2002; Schweinberger & Soukup, 1998; Schweinberger et al., 1999), strongly suggests that the assumption of independence in the processing and representation of identity and expression needs to be systematically examined. A critical issue that arises, however, is the actual *definition* of independence and, by extension, its complement. In all of the work to date, the definitions have been operational and specific to each of the experimental tasks, making it difficult to relate the definitions. Consequently, we chose to approach the issue using two theoretical characterizations of independence—general recognition theory and systems factorial theory—along with the traditional operational definitions and converging operations central to the notion of separability in the Garner approach (Garner, Hake, & Eriksen, 1956; Garner & Morton, 1969).

Separability From the Perspective of Garner's Speeded Classification Task

Garner's speeded classification task (Garner, 1974, 1978; Garner & Felfoldy, 1970) has been used as a test of whether or not two dimensions of a stimulus (e.g., identity and expression) are processed independently. Observers in this task are asked to classify stimuli under two conditions. In the *baseline* condition, the task-relevant dimension (e.g., expression) is varied while the task irrelevant dimension (e.g., identity) is held constant. In the *filtering* condition, values of the task-irrelevant dimension are allowed to vary from trial-to-trial. Performance in these two conditions is then compared using mean RTs. If the task-irrelevant variation in the filtering condition slows performance with the target dimension (relative to the baseline condition), then the dimensions are inferred to be *integral.* If, on the other hand, performance in the filtering and baseline condition is equivalent, the dimensions are inferred to be *separable.* The magnitude of the difference in mean

[2] A reviewer commented that a close look at the article by Hasselmo et al. (1989) will reveal that the findings do provide evidence for a small number of neurons that exhibit combined sensitivity to facial expression and identity.

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RTs between the baseline and filtering conditions is referred to as the level of *Garner interference* (GI).

Independence and Separability From the Perspective of General Recognition Theory

Although the Garner logic is well established, it is, at core, operational and thus subject to all of the limitations of operational definitions (Maddox, 1992). In addition, it is not clear in those operational definitions whether the construct of separability is intended to apply to encoded representations or the processes that operate on them. An alternative to characterizing separability and independence is available in the multidimensional generalization of signal detection theory known as general recognition theory (GRT, Ashby & Townsend, 1986; Kadlec, 1992; Kadlec & Townsend, 1992b; Thomas, 1995, 2001a). GRT provides a metatheoretical language for representing hypotheses regarding the presence or absence of interactions among the dimensions of encoded information. GRT distinguishes three ways that, separately and in combination, independence and its logical alternatives can be characterized. A point to emphasize here is that these characterizations start with an intentionally general theoretical abstraction (specifically, the parameters of the multivariate probability distributions on the levels of encoded information, and the parameters of the decision boundaries) and work to the predicted behavioral patterns.

The encoded sources of perceptual information about each of the two stimulus attributes are said to be *perceptually independent* (PI) if the encoded representations of those two attributes are stochastically (statistically) independent.[3] PI is the only "flavor" of independence in GRT that applies at the level of the individual stimulus. Thus, violations of PI have been suggested as the strongest representation of dimensional interactions within GRT (see O'Toole, Wenger, & Townsend, 2001; Wenger & Ingvalson, 2002, 2003, for discussions pertinent to face perception). The second of the two GRT constructs is known as *perceptual separability* (PS). If the two sources of perceptual information in the encoded representation are perceptually separable, then the effect of a dimension at one level does not depend on the level of the other dimension. Because PS is defined in terms of effects across stimuli, it is the GRT construct closest to the Garnerian notion of separability (Maddox, 1992). The final construct of GRT is *decisional separability* (DS): as it is defined in GRT, decisional separability of the component dimensions is obtained when the decision rule (i.e., the decision bound) for one dimension is unaffected by the level of the other dimension.[4]

Independence From the Perspective of Systems Factorial Technology

Where GRT can be understood to address the question of independence at the level of internal representations, systems factorial technology (SFT) addresses this question at the level of the processes that operate on those representations. SFT is the result of more than three decades of work (Schweickert, 1978, 1983; Schweickert, Giorgini, & Dzhafarov, 2000; Schweickert & Townsend, 1989; Townsend, 1972, 1974; Townsend & Ashby, 1978; Townsend & Nozawa, 1995; Townsend & Wenger, 2004a, 2004b) characterizing four fundamental aspects of real-time information processing: *architecture, stopping rule, capacity and independence*. These aspects can be used to characterize any information-processing system needing to process two or more sources of information. With respect to the present study, a critical point to emphasize is that SFT is concerned explicitly with independence in the *rate* with which the information sources are processed, not in the distribution of encoded information.

To illustrate the characterization of independence from the perspective of SFT, consider again the example of identity and expression (Ganel & Goshen-Gottstein, 2002, 2004). If the rate at which expression is processed does not influence the rate at which identity is processed, then the two channels are considered independent. In contrast, if the rate at which expression is processed influences the rate for identity, then the two channels are considered dependent in rate. Both positive and negative dependencies can exist (Wenger & Townsend, 2006). However, as Townsend has forcefully argued (Townsend, 1974; Townsend & Ashby, 1983; Townsend & Thomas, 1994; Townsend & Wenger, 2004a, 2004b), the question of independence in rate, while logically independent of questions of architecture, stopping rule, and capacity, cannot be theoretically characterized or empirically decided without consideration of these three other aspects of real-time processing. Consequently, the theoretical analysis of the central empirical hypothesis for the current work will of necessity consider all four characteristics of processing, and not just independence.

Summary and Predictions

We consider the parallel route hypothesis (Bruce & Young, 1986) as a set of hypotheses framed in the terminology of Garner's speeded classification task, and the GRT and SFT constructs (see also Ingvalson & Wenger, 2005; O'Toole et al., 2001). As a first-order prediction, then, to the extent that the three perspectives on independence or separability (or, more critically, violations of independence or separability) are concerned with the same underlying construct, the Garner, SFT, and GRT perspectives on independence lead to nine unique hypotheses. Table 1 gives the summary of the hypotheses. Hypotheses 1(a) and 1(b) are framed in the language of Garner's speeded classification task. Hypotheses 2(a)–2(c) use GRT constructs. Hypotheses 3(a)–3(d) translate the parallel route hypothesis to SFT concepts.

Hypothesis 1(a) assumes that identity and expression are *separable dimensions,* from the perspective of the Garner speeded-

[3] We here provide general verbal definitions of each of the GRT constructs. However, we emphasize that these constructs are completely and explicitly formalized in GRT. These formal definitions can be found in a range of publications that describe work with GRT, including the seminal paper (Ashby & Townsend, 1986).

[4] Note, independence at the level of the individual stimulus, as captured by PI, requires a separate testing for each stimuli in the set, with each dimension in a given stimulus retaining its unique (single) value. Independence at a across stimuli level, as captured by PS, DS, and Garner interference, is tested with respect to the entire ensemble of stimuli in the experimental set. In these tests, dimensions are generalized over stimuli and can take all possible values. This form of independence measures the influence of variation in one dimension on performance with the other dimension. (Ashby & Townsend, 1986).

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Table 1
Summary of the Parallel-Route Hypotheses Regarding Separability and Independence From the Perspectives of the Garner Tradition, GRT, and SFT

Expt.	Perspective	Hypothesis number	Predicted inference
1	Garner	1 (a)	Separability (no GI)
1	Garner	1 (b)	Separability (no RG)
2	GRT	2 (a)	Perceptual independence
2	GRT	2 (b)	Perceptual separability
2	GRT	2 (c)	Decisional separability
3	SFT	3 (a)	Parallel architecture
3	SFT	3 (b)	Self-terminating or exhaustive stopping rule
3	SFT	3 (c)	Unlimited capacity
3	SFT	3 (d)	Independence in rate

Note. Expt. = Experiment; GI = Garner interference; RG = Redundancy gains.

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classification task. Consequently, identity and expression should produce no GI. Although this hypothesis has been tested (and invalidated) in previous studies (Ganel & Goshen-Gottstein, 2004; Schweinberger et al., 1999; Schweinberger & Soukup, 1998; Le-Gal & Bruce, 2002), it serves us as a point of connection to the literature, by allowing us to demonstrate that our stimuli produce results that are comparable to those of (e.g.) Ganel and Goshen (2004). Hypothesis 1(b) approaches this same question from the perspective of facilitative rather than inhibitory interactions. Specifically, if separability holds there should be no improvements in performance attributable to experimental correlations of identity and expression.

Hypotheses 2(a)–2(c) refer to tests of independence and separability from the perspective of GRT. Hypothesis 2(a) suggests that if identity and expression are encoded independently for any one stimulus, then the distributions of perceptual evidence for these two dimensions should be statistically independent of one another, and PI should hold. Hypothesis 2(b) suggests that if variations in identity have no effect on the distribution of perceptual evidence for expression (and vice versa), then the two dimensions should be perceptually separable (i.e., PS should hold). Hypothesis 2(c) suggests that if the variations in identity have no effect on the manner in which observers make decisions about expression (and vice versa), then the decisional criteria for each dimension should be invariant across levels of the other dimension and DS should hold.

Hypotheses 3(a)–3(d) translate the assumptions of the parallel route hypothesis to a set of assumptions regarding the four general characteristics of information processing considered by SFT (see Ingvalson & Wenger, 2005, for a closely related effort). With respect to the goals of the present study, the predictions regarding independence in rate (hypothesis 3(d)) are most critical. However, as noted above, inferences regarding any one of these dimensions has to be understood in the context of inferences regarding the other three (Townsend & Nozawa, 1995; Townsend & Wenger, 2004a, 2004b); consequently, we consider the mapping of the assumptions of the parallel-route hypotheses to all four characteristics of processing.

Hypothesis 3(a) adopts the general assumption of parallel processing of identity and expression (as in Haxby et al., 2000, 2002). Hypothesis 3(b) assumes that the stopping rule is task-dependent and as such will be either self-terminating or exhaustive, depending on the response rule used in the experimental task. Hypothesis 3(c) suggests that if identity and expression do not influence each other at the level of processing, then adding the requirement to process identity should not impair the ability to process expression (and vice versa); as such, processing should be of unlimited capacity. For a parallel processing architecture, should the channels for identity and expression interact in either inhibitory or facilitative ways (as would be the case for interference or facilitation in the Garner task), then processing should be either limited capacity or supercapacity (see also O'Toole et al., 2001; Wenger & Townsend, 2006). Finally, Hypothesis 3(d) suggests that the channels processing identity and expression should evidence independence in rates of processing.

A critical aspect of our approach to testing the hypothesis of independence from three perspectives is our ability to examine the extent to which the three conceptions are capturing a common underlying construct. As noted earlier, the Garnerian concept of separability and the GRT construct of PS are consistent with one another, in the sense that both address changes in performance across variations in the levels of the stimulus dimensions (see also Maddox, 1992). However, to our knowledge, there has been no systematic attempt to consider this range of characterizations of the construct of independence within the performance of individual observers, using a common set of stimuli. Thus, one critical contribution of the present effort is both the comprehensiveness of the set of characterizations and the ability to relate inferences across that set within the performance of individual observers (Ashby, Maddox, & Lee, 1994; Busemeyer & Diederich, 2010; Estes, 1956; Estes & Maddox, 2005; Fitousi & Wenger, 2011; Maddox, 1999).

Overview of the Experiments

Three experiments were performed to test and contrast within the performance of individual observers the set of hypotheses presented in Table 1. Each experiment addressed a set of hypotheses associated with one of the approaches (Garner, GRT, or SFT). In Experiment 1 we applied Garner's speeded classification task to the dimensions of identity and expression, using both familiar and unfamiliar faces. This experiment was designed as partial replication of Ganel and Goshen-Gottstein's (2004) Experiment 2. Once we establish that our stimuli are capable of supporting a replication

of the results obtained by Ganel and Goshen-Gottstein (2004), we can then proceed to examine performance with these stimuli in the remaining two experiments, using a complete identification paradigm in Experiment 2 to test the predictions regarding PI, PS, and DS, then using a double-factorial paradigm in Experiment 3 to test the prediction regarding independence in rate, along with architecture, stopping rule, and capacity.

Experiment 1

The critical first step in this work was to replicate the core results of Ganel and Goshen-Gottstein's (2004) Experiment 2. These researchers argued that, in contrast to the parallel route hypothesis (Bruce & Young, 1986), identity and expression interact. Based on previous work by Schweinberger and colleagues (Schweinberger et al., 1999; Schweinberger & Soukup, 1998), these researchers assumed that if relative discriminability is controlled appropriately, and the dimensions are tested in Garner's speeded classification task, they should appear as *integral dimensions* (e.g., produce reliable levels of symmetric GI).

In addition, Ganel and Goshen-Gottstein hypothesized that familiarity with the presented faces should modulate the strength of the observed interaction. In agreement with their conjecture, Ganel and Goshen-Gottstein found higher levels of cross-talk (as measured by GI) between identity and expression for familiar than for unfamiliar faces. In Experiment 1 we intentionally controlled for relative disriminabilty. Following the lead of Ganel and Goshen-Gottstein (2004), we also manipulated familiarity. In addition, we collected a sufficient amount of data to allow for statistical analysis at the level of the individual observer.

Method

Participants. Four participants were recruited from the student population of the Pennsylvania State University. Participants were reimbursed for their participation at the rate of $8/hr. All participants reported normal or corrected-to-normal vision, normal hearing, and unencumbered use of both hands. Each observer completed four 1-h sessions across four consecutive days.

Materials. Stimuli consisted of eight gray-scale images, resulting from the crossing of two identities and two expressions, at two levels of familiarity (familiar, unfamiliar). Pictures of two contemporarily well-known actresses—Natalie Portman and Keira Knightley—were used as the familiar stimuli (Figure 1a). Pictures of two undergraduate research assistants in our laboratory were used as the unfamiliar stimuli (Figure 1b). Two levels of expression (e.g., happy and neutral) were used for each identity at each level of familiarity. For the purposes of Experiment 1, it was important to make identity and expression equally discriminable. This was achieved by using faces that are very similar (Ganel & Goshen-Gottstein, 2004). The relative discriminability of identity and expression was validated in pilot work. The images were equated on brightness, size, and shape. Viewed at a fixed distance of 76 cm, the images subtended 2.6° of visual angle horizontally and 3.2° vertically. To exclude the possibility that observers will

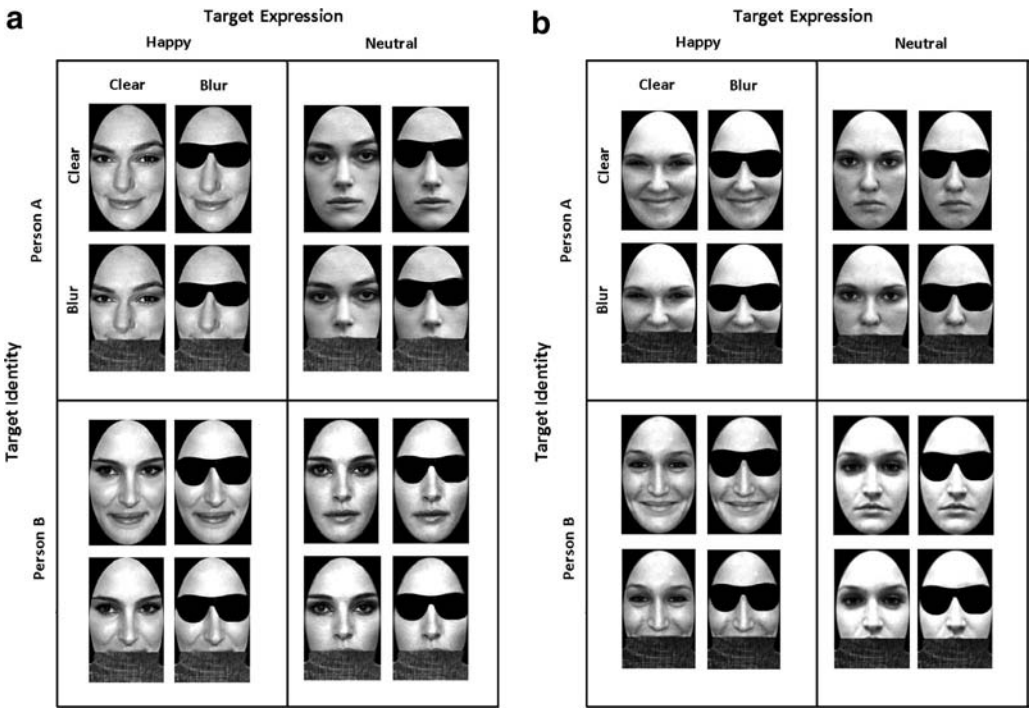


Figure 1. (a) Familiar and (b) unfamiliar stimuli used in the experiments. The pictures of Natalie Portman and Keira Knightley were used as familiar faces. The pictures of two undergraduate research assistants were used as unfamiliar faces. Sun glasses and scarves were used in Experiment 3 to selectively influence the speed of processing of identity and expression. For Experiments 1 and 2, only the top left face from each group was used. The role of sweater and sunglasses will be discussed in the relevant sections on experimental method.

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use information coming from the different shapes of faces, all the faces were presented with the same oval shape.

Stimuli were presented and responses were recorded using a PC-compatible computer with a 19″ (diagonal) color CRT set to a resolution of 1024 × 768 pixels. Stimulus onsets were synchronized with the vertical retrace of the monitor (refresh rate of 100 Hz), and responses were timed to ± 1 ms using Macromedia's Authorware software (Macromedia, 2007). Observers viewed the display at a fixed distance of 76 cm through a black viewing tube, used to restrict the light available to observers' eyes to only the light emitted by the display. One end of the viewing tube was placed against the display, while the other was placed against the observer's chin rest.

Design and procedure. Two tasks were used in Experiment 1, one requiring identification of expression and the other requiring selection of identity. The experiment was designed as a 2 (familiarity: familiar, unfamiliar) × 2 (task: identity, expression) × 4 (condition: 2 baseline, positively correlated, negatively correlated, filtering) complete factorial design, with all factors manipulated within subjects. In baseline blocks, the dimension of interest (the levels of the relevant dimension varied from trial to trial, while the other (irrelevant) dimension was held fixed. In analysis, performance measures in the two baseline blocks were averaged (see, e.g., Fitousi, Shaki, & Algom, 2009). In filtering blocks, the two dimensions varied orthogonally, and observers were required to respond to only one of the dimensions. The two blocks of correlated trials varied in terms of the sign of the correlation. In the positively correlated block, levels of the two dimensions always occurred together (e.g., Keira Knightley appeared always with a happy expression whereas Natalie Portman was shown in a neutral face), while in negatively correlated blocks, the opposite was true. For each relevant dimension there was a sequence of filtering, two correlated blocks (positive and negative) and two baseline blocks. The order of the blocks within a sequence was randomly assigned under the restriction that blocks of the same type (i.e., baseline, correlated) are presented sequentially, but their order can be random. Each baseline and correlated block consisted of 27 trials. Each filtering block consisted of 54 trials. The critical dependent measures were (a) the difference in mean RT between the filtering and baseline conditions (Garner interference, GI), and (b) the difference between the mean RTs for the baseline and the positively and the negatively correlated blocks.

In the expression task, participants had to classify faces as either happy or neutral. Participants pressed the right-hand key ("m") if the expression was judged to be happy and a left-hand key ("z") if the expression was judged to be neutral. When judging identity, observers responded by pressing a right-hand key if the face was judged to be person A and a left-hand key if the face was judged to be person B. The keys were assigned such that each participant responded with the same response assignment across the course of the entire experiment.

Before testing any participant, we verified that they could quickly and accurately name the familiar faces. To do this, participants were presented with the pictures of the two famous individuals. Participants were asked to name the faces and add semantic details (movies they have acted in etc.) about the two persons. Participants who could not meet this criteria were excluded from the experiment. Participants who met this criteria began by receiving identification practice on the eight faces. Participants were presented with two blocks of 100 practice trials each. Each block consisted of either four familiar or four unfamiliar faces. The sequence of the blocks was determined randomly, under the constraint that each block appeared an equal number of times for each observer. On each trial, one face was presented, with order of presentation randomized for each participant. The participant was instructed to respond to the face on the screen according to the following response assignment: "1" for Person A, smiling, "2" for person A, neutral, "9" for person B, smiling, and "0" for person B, neutral. To facilitate learning, the name, expression, and response-key related to the presented face were written on the screen in Times New Roman font size 48. At the end of each practice block, participants were tested on 20 additional trials. Participants had to score at least 18 correct responses in that condition to proceed to the experimental blocks. All observers met this criteria.

Each experimental block started with a fixation cross appearing on the center of the screen for 100 ms, followed by black screen for another 100 ms. Then, a gray-scale image of one of the faces was presented until the observer provided a response. This was followed by a black screen for 500 ms. The sequence of the baseline, correlation, and filtering blocks was determined randomly, under the constraint that baseline blocks always appeared one after the other (in random order). The same constraint applied to correlation blocks.

Results

Data were analyzed with the goal of testing hypotheses 1(a) and 1(b), at both the aggregate and individual levels (see, Table 1). According to these hypotheses, identity and expression do not interact and thus should appear as separable.

Two traditional diagnostics of separability were used: Garner interference (GI) and redundancy gains (RG) (see for, Pansky & Algom, 1999; Garner, 1974; Melara & Marks, 1990a, 1990b; Pomerantz, 1986). GI is computed as a measure of difference between performance in filtering, where the two dimensions (identity and expression) are varied orthogonally, and baseline, where one dimension is held fixed at one level. Parity implies that observers were able to attend to the criterial dimension (e.g., identity) without being distracted by irrelevant variation on another dimension (e.g., expression). Conversely, if observers' performance at classifying values on the relevant dimension is better in baseline than at filtering (i.e., if reliable levels of GI are observed), then selective attention has been compromised. Hypothesis 1(a) predicts that no GI should appear when classifying either identity or expression.

Another test that distinguishes integral and separable dimensions uses performance in the correlated blocks (Garner, 1974; Pomerantz, 1986). Redundancy gains (RG)[5] gauge differences in RT between baseline and positively correlated blocks or/and baseline and negatively correlated blocks. If a pair of dimensions is integral, then classification in the correlated tasks is better than that at baseline. For such dimensions, observers obtain a benefit from correlated values on an irrelevant dimension and suffer GI from an

[5] The term redundancy gains (e.g., RG) used in this study is distinct from the use of this term in typical redundant-target tasks (e.g., Miller, 1982; Townsend & Nozawa, 1995).

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orthogonal variation. If a pair of dimensions is separable, observers do not benefit from the correlated blocks nor exhibit GI. Hypothesis 1(b) predicts that observers should not benefit from the correlation, indicating that the dimensions are separable.

Statistically, we assessed GI and RG separately in three different analyses of variance (ANOVAs), with the task's independent variable restricted to the two levels pertinent to the measure of interest. Specifically, to estimate GI and its interaction with familiarity, we restricted our consideration to the baseline and filtering conditions. To estimate the presence of RG in positively correlated blocks, we used only the baseline and positively correlated blocks.

In all analyses, RTs shorter than 150 ms or longer than 2,000 ms were excluded from analysis (a total of 2% of all trials across all observers). Error rates averaged 8% across all observers, and RTs for errors were excluded from the analyses. An α-level of 0.05 was used throughout. The presentation order of our results corresponds to the hypotheses: thus, we first report on the GI and then on the RG results. Within each section we start with the aggregate analysis and then proceed to present the individual level results.

Before presenting the main results, we tested whether the dimensions of identity and expression matched on discriminability, within- and across levels of familiarity. To do so, we compared performance at baseline blocks for each dimension—a conventional index for the assessment of relative discriminablity (Algom, Dekel, & Pansky, 1996; Pansky & Algom, 1999; Melara & Mounts, 1993). Tables 2 and 3 present the mean RT and error rates per condition and observer, for both familiar and unfamiliar faces. As can be noted, identity and expression were classified at roughly the same speed in the baseline conditions (460 msec vs. 485 msec, for familiar faces, and 458 msec vs. 463 msec, for unfamiliar faces). This pattern, which has been observed for accuracies as well, suggests that identity and expression are equally discriminable. In this case, Garner interferences cannot be ascribed to an imbalance in discriminablity (Garner & Felfoldy, 1970; Melara & Algom, 2003). Note that across dimensions, the two levels of familiarity were processed similiraliy (473 msec vs. 460 msec), pointing to matched discriminablity of familiar and unfamiliar sets of faces as well.

Garner interference. RTs for correct responses were analyzed using a three-way analysis of variance (ANOVA), with familiarity (familiar, unfamiliar), dimension (identity, expression), and task (baseline, filtering) as factors. The baseline mean RT was an average of mean RTs in the two types of baseline trials. This analysis was performed twice: once at the level of the aggregate and once at the level of the individual observers. The results of these analyzes are presented in Table 4.

We start by reporting the results of the aggregate analyses. The most important outcome is the presence of GI, as captured by a reliable main effect of task. This indicates that performance was better in the baseline (486 ms) relative to the filtering condition (585 ms), an outcome that is consistent with previous studies that found symmetric GI in judgments of both identity and expression (Ganel & Goshen-Gottstein, 2002, 2004). A task $\times$ dimension interaction revealed that GI levels were modulated by the dimension to which observers attended. Specifically, larger levels of GI were observed in classification of identity than in judgments of expression (differences in the means of 115 ms and 84 ms, respectively). In contrast to Ganel and Goshen-Gottstein's hypothesis, familiarity had no impact on the magnitude of GI, as evidenced by a familiarity $\times$ task interaction that failed to reach criterion for reliability.

The results reported so far are averages across observers, which may or may not be reflective of regularities at the level of individual observers (Estes, 1956; Fitousi & Wenger, 2011; Maddox, 1999). Consequently, we repeated the same analyses at the level of the individual observers, with the results presented in the lower portion of Table 4. High level of consistency has been achieved between the aggregate and the individual results.

To estimate the magnitude of GI, we conducted a series of Tukey's tests, using the data from the baseline and filtering conditions. Figure 2 plots the mean level of GI for each observer, along with the corresponding standard errors (obtained from the Tukey's tests), for judgments on each of the two dimensions. As can be seen, appreciable levels of reliable GI were obtained for both type of judgments for all four observers. In addition, the magnitude of GI was larger for identity judgments than for classification of expression. Excepting the failure to obtain Ganel and Goshen-Gottstein's (2004) modulation of GI by familiarity, our data represent a general replication of Ganel and Goshen-Gottstein (2004), at both the aggregate an individual levels.

Table 2
Experiment 1: Summary of Mean RTs (ms) for the Garner Speeded Task

		Familiar					Unfamiliar				
Dim	Obs.	O1	O2	O3	O4	A	O1	O2	O3	O4	A
E	B	397	396	590	458	460	404	377	593	460	458
	PC	411	397	565	464	459	425	391	580	476	468
	NC	403	398	616	459	469	406	372	530	454	440
	F	534	477	632	560	550	546	484	628	555	553
	GI	137	81	42	102	90	142	107	35	95	95
I	B	390	414	643	493	485	405	382	590	475	463
	PC	440	459	645	528	518	435	427	602	480	486
	NC	400	392	584	480	464	423	395	576	444	459
	F	567	596	688	657	627	581	546	677	629	608
	GI	177	182	45	164	142	176	164	87	154	145

Note. Dim = Dimension; E = Expression; I = Identity; O = Observer; A = Aggregate; B = Baseline; PC = Positively correlated; NC = Negatively correlated; F = Filtering; GI = Garner interference (F – B).

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Table 3
Experiment 1: Summary of Error Rates P(Correct) for the Garner Speeded Task

		Familiar					Unfamiliar				
Dim	Obs.	O1	O2	O3	O4	A	O1	O2	O3	O4	A
E	B	0.90	0.96	0.97	0.95	0.95	0.90	0.96	0.97	0.93	0.95
	PC	0.92	0.95	0.96	0.94	0.94	0.92	0.96	0.96	0.90	0.94
	NC	0.94	0.96	0.99	0.92	0.95	0.94	0.96	0.99	0.95	0.96
	F	0.87	0.93	0.93	0.90	0.90	0.88	0.93	0.94	0.88	0.90
	GI	0.03	0.03	0.04	0.05	0.05	0.02	0.00	0.03	0.05	0.05
I	B	0.90	0.95	0.95	0.94	0.94	0.86	0.95	0.95	0.94	0.93
	PC	0.90	0.96	0.95	0.90	0.93	0.90	0.96	0.95	0.90	0.93
	NC	0.91	0.95	0.98	0.95	0.95	0.91	0.96	0.98	0.95	0.95
	F	0.85	0.92	0.93	0.88	0.90	0.85	0.93	0.93	0.89	0.90
	GI	0.05	0.03	0.02	0.06	0.04	0.01	0.02	0.02	0.05	0.03

Note. Dim = Dimension; E = Expression; I = Identity; O = Observer; A = Aggregate; B = Baseline; PC = Positively correlated; NC = Negatively correlated; F = Filtering; GI = Garner Interference (B – F).

Redundancy gains with positively correlated blocks. RTs for correct responses were next analyzed using a three-way ANOVA, with familiarity (familiar, unfamiliar), dimension (identity, expression), and task (baseline, positively correlated) as factors, first at the level of the aggregate and then for each observer separately. Results of these analyses are presented in Table 5.

At the level of the aggregate results, the most critical outcome is the presence of RG, as captured by a reliable main effect of task. Performance was better in the positively correlated block (459 ms) relative to the baseline block (486 ms), an outcome that is novel to the present effort, as previous studies did not use the correlated condition (Ganel & Goshen-Gottstein, 2002) or used it to address a different question than ours (Ganel & Goshen-Gottstein, 2004). A marginally reliable ($p = .07$) task × dimension interaction revealed that RG levels were modulated by the dimension to which observers attended. Specifically, larger RGs were observed in classification of identity than in judgments of expression (differences in the means of 40 ms and 15 ms, respectively). Familiarity had no impact on the magnitude of RG, as evidenced by a familiarity × task interaction that failed to reach criterion for reliability.

We repeated the same analyses at the level of the individual observers, with the same level of consistency. The results are presented in the lower portion of Table 5. Figure 3 plots the mean level of RG for each observer, along with the corresponding standard errors (derived from the Tukey's tests), for judgments on each of the two dimensions. Overall, the individual results conform with the aggregate.

Redundancy gains with negatively correlated blocks. RTs for correct responses were next analyzed using a three-way

Table 4
Experiment 1: ANOVA on RTs for Correct Responses, for the Aggregate (A), and Separately for Each of the Observers (O)

O	Effect	*df*	*MSE*	*F*	O	Effect	*df*	*MSE*	*F*
A	F	1, 3	370.44	2.89					
	D	1, 3	700.19	28.92**					
	T	3, 9	2290.36	35.15‡					
	F × D	1, 3	137.53	11.10**					
	F × T	3, 9	48.61	1.94					
	D × T	3, 9	529.09	10.64†					
	F × D × T	3, 9	112.36	0.72					
1	F	1	131640.70	3.81*	2	F	1	456255.65	18.12‡
	D	1	543226.97	15.74‡		D	1	5710997.67	226.87‡
	T	1	210156929.16	608.78‡		T	1	13530421.42	537.49‡
	F × D	1	1514.32	0.04		F × D	1	501274.70	19.91†
	F × T	1	114.38	0.00		F × T	1	50	0.08
	D × T	1	270743.36	7.84†		D × T	1	510656.54	20.29‡
	F × D × T	1	10304.42	0.30		F × D × T	1	29605.48	0.80
3	F	1	370801.96	12.34‡	4	F	1	298710.42	6.44†
	D	1	2403977.01	79.99‡		D	1	3997726.14	86.15‡
	T	1	3163969.44	105.28‡		T	1	14339179.16	308.99‡
	F × D	1	111851.69	3.72*		F × D	1	475259.14	10.24‡
	F × T	1	113822.65	3.79*		F × T	1	1262.33	0.03
	D × T	1	67911.35	2.26		D × T	1	749434.58	16.15‡
	F × D × T	1	48281.57	1.61		F × D × T	1	90948.94	1.96

Note. F = Familiarity (familiar, unfamiliar); D = Dimension (identity, expression); T = Task (baseline, filtering).
* $p < .07$. ** $p < .05$. † $p < .01$. ‡ $p < .001$.

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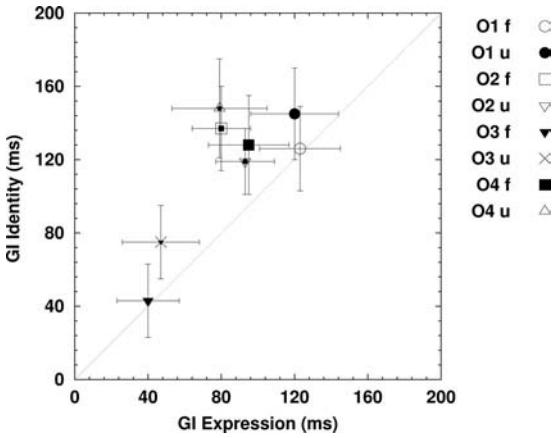


Figure 2. Experiment 1: Garner interference (GI) estimated for each observer (O), for familiar (f) and unfamiliar (u) faces, for judgments of identity and expression.

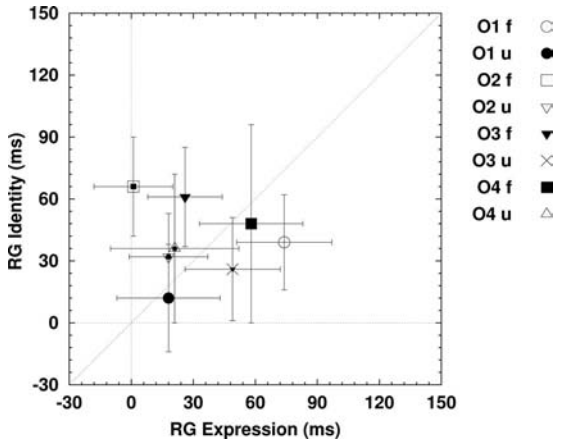


Figure 3. Experiment 1: Redundancy gains (RG) as captured by Tukey's tests. The tests were performed on positively correlated blocks and baseline blocks, for each observer (O), for familiar (f) and unfamiliar (u) faces, and criterial dimension (identity or expression).

ANOVA, with familiarity (familiar, unfamiliar), dimension (identity, expression), and task (baseline, negatively correlated) as factors, first at the level of the aggregate and then at the level of the individual observers. Table 6 presents the results of these analyses.

At the level of the aggregate, the most important result is the presence of reliable RGs, as reflected by a reliable main effect of task. Performance was better in the negatively correlated trials (470 ms) than in the baseline trials (486 ms). A significant task × dimension interaction revealed that RG levels were modulated by the dimension to which observers attended. Specifically, larger RGs were observed in classification of identity than in judgments of expression (differences in the means of 29 ms and 4 ms, respectively). All the other effects failed to reach criterion for statistical reliability.

We repeated the same analyses at the level of the individual observers, with the results presented in the lower portion of Table 6. Figure 4 plots the mean level of RG for each observer, along with the corresponding standard errors bars (derived from the Tukey's tests), for judgments on each of the two dimensions.

Discussion

Experiment 1 was intended to provide an empirical foundation for the experiments that we report in the pages that follow, by

Table 5
Experiment 1: ANOVA on RTs for Correct Responses, for the Aggregate (A), and Separately for Each of the Observers (O)

O	Effect	*df*	*MSE*	*F*	O	Effect	*df*	*MSE*	*F*
A	F	1, 3	440.45	1.51					
	D	1, 3	337.33	37.00‡					
	T	3, 9	3860.50	104.66‡					
	F × D	1, 3	523.00	0.29					
	F × T	3, 9	148.00	0.16					
	D × T	3, 9	529.09	1666.37*					
	F × D × T	3, 9	589.37	2.69					
1	F	1	47982.14	1.65	2	F	1	178887.56	7.40‡
	D	1	190216.47	6.55**		D	1	1022168.08	42.28‡
	T	1	264504.13	9.10‡		T	1	666202.44	27.56‡
	F × D	1	7810.48	0.27		F × D	1	8965.92	0.37
	F × T	1	10921.53	0.38		F × T	1	6753.69	0.28
	D × T	1	28319.58	0.97		D × T	1	341704.92	14.14‡
	F × D × T	1	66635.09	2.29		F × D × T	1	133746.37	5.53**
3	F	1	512657.06	22.04‡	4	F	1	307480.61	7.29‡
	D	1	1090230.43	46.88‡		D	1	523753.40	12.42‡
	T	1	1322023.22	56.84‡		T	1	602615.32	14.29‡
	F × D	1	34586.34	1.49		F × D	1	536764.76	12.73‡
	F × T	1	8668.65	0.37		F × T	1	432.17	0.01
	D × T	1	7708.41	0.33		D × T	1	159904.12	3.79**
	F × D × T	1	169615.14	7.29‡		F × D × T	1	37461.66	0.89

Note. F = Familiarity (familiar, unfamiliar); D = Dimension (identity, expression); T = Task (baseline, positively correlated).
* $p < .07$. ** $p < .05$. † $p < .01$. ‡ $p < .001$.

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Table 6
Experiment 1: ANOVA on RTs for Correct Responses, for the Aggregate (A), and Separately for Each of the Observers (O)

O	Effect	*df*	*MSE*	*F*	O	Effect	*df*	*MSE*	*F*
A	F	1, 3	548.87	1.97					
	D	1, 3	669.91	17.70					
	T	3, 9	4777.91	22.72**					
	F × D	1, 3	138.25	4.43					
	F × T	3, 9	137.75	0.71					
	D × T	3, 9	529.09	189.70**					
	F × D × T	3, 9	95.20	0.30					
1	F	1	38503.49	1.44	2	F	1	396342.40	16.30‡
	D	1	110787.91	4.15**		D	1	1178333.94	48.45‡
	T	1	591601.56	22.18‡		T	1	529651.34	21.78‡
	F × D	1	19437.39	0.73		F × D	1	100241.77	4.12**
	F × T	1	6188.046	0.23		F × T	1	9767.68	0.40
	D × T	1	90513.40	3.39*		D × T	1	279911.13	11.51‡
	F × D × T	1	34761.25	1.30		F × D × T	1	9118.41	0.37
3	F	1	84080.62	30.63‡	4	F	1	179303.33	4.22**
	D	1	749791.26	27.17‡		D	1	845971.62	19.89‡
	T	1	24660.08	0.89		T	1	180307.82	4.24**
	F × D	1	216192.11	7.83†		F × D	1	454146.91	10.68‡
	F × T	1	22013.46	0.80		F × T	1	19852.72	0.47
	D × T	1	133506.74	4.84**		D × T	1	19852.72	0.47
	F × D × T	1	399.18	0.01‡		F × D × T	1	74028.23	1.74

Note. F = Familiarity (familiar, unfamiliar); D = Dimension (identity, expression); T = Task (baseline, negatively correlated).
$^{*}p < .07$. $^{**}p < .05$. $^{†}p < .01$. $^{‡}p < .001$.

providing a link between our efforts and those in the literature (Ganel & Goshen-Gottstein, 2004, in particular). In contrast to Hypothesis 1(a) we found appreciable amounts of GI in the classification of both dimensions. Identity interfered with judgments of expression, whereas expression intruded on classifications of identity. Larger levels of GI were observed in the latter than in the former. Hypothesis 1(b) tested a complementary assertion, specifically, that if identity and expression are separable dimensions, no RG should be observed. We found appreciable levels of RG for both positively and negatively correlated blocks, at the aggregate level. The evidence for RG at the individual level was present but evidenced individual differences. These outcomes are inconsistent with the predictions of the parallel-route hypothesis, thus leading to a rejection of hypothesis 1(b).

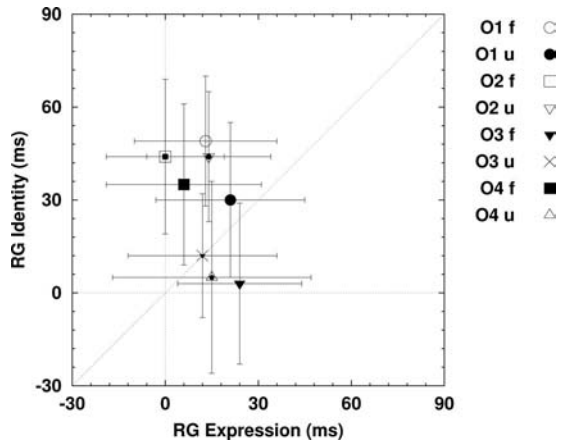


Figure 4. Experiment 1: Redundancy gains (RG) in negatively correlated blocks, for each observer (O), for familiar (f) and unfamiliar (u) faces, and criterial dimension (identity or expression).

Experiment 2

The results of Experiment 1 suggest that identity and expression are processed as integral rather than separable dimensions, contradicting the independence assumption of the parallel-route hypothesis. Experiment 2 was intended to explore this interaction of dimensions from the perspective of GRT, which allows within- and across-stimulus forms of independence and separability to be distinguished, and allows for perceptual and decisional factors to be identified. Experiment 2 tested hypotheses 2(a)–2(c) (see Table 1), framed in terms of three characterizations of independence and separability. Specifically, if identity and expression are evaluated independently, then the performance of observers should be consistent with perceptual independence (hypothesis 2(a)), perceptual separability (hypothesis 2(b)), and decisional separability (hypothesis 2(c)). In contrast, given that integrality from the Garnerian perspective is most consistent with a violation of perceptual separability (Ashby & Maddox, 1994; Maddox, 1992), the integrality revealed in the data from Experiment 1 suggests that we should observe violations of PS in Experiment 2. In considering this possibility, it should be emphasized that Experiment 2 involved the same stimuli and the same observers as Experiment 1.

Method

Participants. The same four observers who participated in Experiment 1 served as observers in Experiment 2. As in Experiment 1, they were compensated at the rate of $8 per hour. Each observer completed four 1-h sessions across four consecutive days.

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Stimuli, apparatus, and procedure. The stimuli and apparatus used in Experiment 2 were the same as those used in Experiment 1. The only difference between the two experiments was in the nature of the experimental task. Observers in Experiment 2 performed a complete identification task in which they were required to judge both the identity and the expression of a stimulus face. Observers performed this task with both the familiar and unfamiliar faces, within separate blocks of trials. Each block consisted of 80 trials. Stimuli at each combined level of identity and expression were presented with equal frequency in each block. Each experimental session consisted of five blocks of familiar faces and five blocks of unfamiliar faces. Half of the sessions for each observer began with a block of familiar faces and half began with a block of unfamiliar faces.

Each trial in each block began with a fixation point appearing at the center of the screen for 300 ms. This was replaced by a target face, whose presentation duration was set separately for each observer in each block, to obtain an overall accuracy level of between 75% and 85%. Each observer started with an exposure duration of 80 msec. This value was updated from block to block according to the following rule: if performance level was above 85% then exposure duration was decreased by 5 msec; if it was below 65% exposure duration was increased by 5 msec (see Copeland & Wenger, 2006, for a similar strategy). Observers responded using the following assignment of keys on the keys at the top of the computer's keyboard: 1 for person A happy, 2 for person A neutral, 9 for person B happy, 0 for person B neutral. No feedback was given. Before the experimental blocks, observers performed in two practice blocks of 100 response-terminated trials each. In those blocks we presented the stimuli, along with the correct key name. Separate practice blocks were used for familiar and unfamiliar faces. All observers reached a criterion level of performance (95% correct) at the end of those blocks.

Results

A summary identification/confusion matrix for each observer at each level of familiarity provided the basis for all of the analyzes. The approach to drawing inferences regarding PI, PS, and DS taken in Experiment 2 combines two statistical approaches that have been applied in studies using GRT. The first combines a set of signal detection measures of sensitivity and bias with a set of nonparametric tests, all applied at the level of the individual dimensions (i.e., marginal measures). This approach dates to the initial work on GRT (Ashby & Townsend, 1986) along with a set of later developments (Kadlec & Townsend, 1992a, 1992b). The second approach (explained in tutorial fashion by Thomas, 2001a) involves the fitting of a set of parametric models to the identification/confusion data, using a hierarchical inference approach (see Copeland & Wenger, 2006; Cornes, Donnelly, Godwin, & Wenger, 2011, for other examples of its application).

Multidimensional signal detection analyses. The first set of analyses consists of tests of equalities that have been shown to hold under assumptions regarding PI, PS, and DS (Ashby & Townsend, 1986; Kadlec, 1992, 1999; Kadlec & Townsend, 1992b). The specific measures used as the basis for these tests are presented in Table 7. Error bars for these measures were computed using the Galanter and Gourevitch method (Gourevitch & Galanter, 1967). Table 8 summarizes the logic for inferences regarding violations.

Note that there are some cases in which inferences are uncertain (see detailed discussions in, e.g., Kadlec & Townsend, 1992a, 1992b). This potential inferential challenge has been noted in various publications since the seminal work on GRT (Ashby & Townsend, 1986). Our approach addresses this challenge by augmenting the nonparametric measures with a hierarchical model fitting procedure developed by Thomas (2001a) to fit various parametrized GRT models to the identification/confusion matrices of each observer. Details of the model-fitting approach are provided in the Appendix.

Results of the marginal analyses and the test of sampling independence are presented in Table 9, and the inferences made on the basis of both the marginal analyses and the model fits are shown in Table 10. It worth noting the high level of correspondence between the inferences drawn from the marginal analyses and the model fits. In some cases, the model fits resulted in identical inferences to those drawn from the marginal analyzes, or resolved

Table 7
Marginal Measures of Performance Used to Guide Inferences Regarding PI (Perceptual Independence), PS (Perceptual Separability), and DS (Decisional Separability), Considering the Relationship Between the Dimensions of Identity (A) and Expression (B)

Quantity	Definition and description
$d'(A, B_j)$	Marginal sensitivity to the identity when expression is in state j (happy, neutral) $= \frac{1}{\sqrt{2}}\left\{z[P(a_1b_j\|A_1B_j)] - z[P(a_1b_j\|A_2B_j)]\right\}$
$c(A, B_j)$	Marginal criterion for reporting the identity when expression is in state j (happy, neutral) $= -\frac{1}{2\sqrt{2}}\left\{z[P(a_1b_j\|A_1B_j)] + z[P(a_1b_j\|A_2B_j)]\right\}$
MRI	Nonparametric equality testing for the presence of marginal response invariance pertinent to inferences regarding PS and DS $P(a_1b_1\|A_1B_1) + P(a_1b_2\|A_1B_1) = P(a_1b_1\|A_1B_2) + P(a_1b_2\|A_1B_2)$
SI	Nonparametric equality testing for the presence of sampling independence when identity is in state 1 and expression is in state j (happy, neutral), pertinent to inferences regarding PI $P(R_{a_ib_j}\|A_1B_j) = [P(a_1b_1\|A_1B_j) + P(a_1b_2\|A_1B_j)] \times [P(a_1b_1\|A_iB_j) + P(a_2b_1\|A_1B_j)]$

Note. Lowercases letters refer to responses, and upper case letters refer to stimuli. Subscript i = Identity 1, Identity 2. Subscript j = happy, neutral.

Table 8
Logic Relating Marginal Measures of Performance to Inferences Regarding PS and DS

Evidence			Inference	
MRI	Equal marginal d'	Equal marginal c	PS	DS
T	T	T	T	T
T	T	F	T	F
T	F	T	F	T
T	F	F	F	F
F	T	T	T	?
F	T	F	T	F
F	F	T	F	?
F	F	F	F	?

Note. Under evidence, T indicates that the equality in question held, F indicates that the equality in question did not hold. Under inference, T indicates no violation, F indicates a violation. A question mark indicates uncertain inference.

uncertainties inherent in the truth table (see Table 8). In other cases, the model fits were nominally different from the marginal analyzes, specifically suggesting a violation where one was not suggested by the marginal analyzes. However, a closer examination of the parameter estimates in these cases showed that the confidence intervals for these estimates overlapped, meaning that the inference of violation was not supported. This high level of agreement between model fits and marginal analysis has been documented in other studies that have used GRT (Copeland & Wenger, 2006; Cornes et al., 2011; Thomas, 2001a).

We proceed by reporting on the results with respect to the hypotheses for Experiment 2. Hypothesis 2(a) predicts that PI should hold for identity and expression. The analyses pertinent to PI revealed no evidence for violations of PI. In particular, the tests of sampling independence (SI) showed no failures of SI (observers 2, 3, 4), or only a small number of failures (observer 1) of SI. In addition, in no cases did models that allowed for nonzero within-stimulus correlations (violations of PI) provide superior accounts of the data. Note that this result does not represent a contradiction of the results obtained in Experiment 1, as those outcomes speak to the GRT concepts of separability rather than perceptual independence.

Next, consider hypotheses 2(b) and 2(c). These hypotheses predicted that PS and DS should hold for identity and expression. With respect to PS, one observer evidenced performance that was consistent with the prediction, with PS holding in all cases. For the remaining three observers, PS was violated in three of the four cases for each observer. Overall, the number of violations of PS violations was larger for familiar (6) than unfamiliar faces (3). With respect to DS, one observer evidenced a complete lack of violations, consistent with the predictions of hypothesis 2(c). However, the remaining three observers evidenced no more than two violations of DS, thus suggesting that perceptual effects were more pervasive than decisional effects.

Discussion

In Experiment 2, we applied the logic and theoretical definitions of GRT (Ashby & Townsend, 1986) to the questions of independence and separability of identity and expression, using the same set of stimuli and the same set of observers that were used in Experiment 1. Three hypotheses were addressed. Hypothesis 2(a) proposed that if identity and expression are encoded independently for any one face, then the statistical effects of the perceptual evidence should appear independent (i.e., PI should hold). This hypothesis was confirmed. We found evidence suggesting PI for all stimuli and for all observers. PI is considered to be the strongest form of independence as it holds within a single face, whereas other GRT constructs (e.g., PS, DS) hold across faces (O'Toole et al., 2001; Wenger & Ingvalson, 2002, 2003). It is also worth noting that finding an absence of violations of PI here does not stand in contradiction to observing GI and RG in Experiment 1, as those measures are defined across rather than within stimuli, and therefore cannot speak to the construct of PI.

Hypothesis 2(b) assumed that if variations in expression have no effect on the distribution of perceptual evidence for identity (and vice verse), then the two dimensions should be perceptually separable, and PS should hold. This hypothesis was refuted for at least the majority of observers. In particular, for familiar faces we found violations of PS for both expression and identity. Interestingly, for unfamiliar faces we found violations of PS for expression but not for identity. Hypothesis 2(c) proposed that the decision bounds for judgments of expression should be invariant across levels of identity, and that the reveres should hold (e.g., DS). This hypothesis was refuted for the majority of observers, as we found violations of DS for both familiar and unfamiliar faces.

Taken together, the results of Experiment 1 and Experiment 2 provide a clear picture. Identity and expression do not interact within a single face; however, when considered from the perspec-

Table 9
Summary of the Marginal Analyses and the Nonparametric Test of Sampling Independence (SI), for Each of the Four Observers (Obs.)

		Obs. 1				Obs. 2				Obs. 3				Obs. 4			
Stimulus	Comparison	MRI	d'	c	SI	MRI	d'	c	SI	MRI	d'	c	SI	MRI	d'	c	SI
Familiar faces	Exp × ide	T	T	F	0	F	F	T	0	T	F	T	0	F	F	T	0
	Ide × exp	T	T	T	0	F	F	T	0	F	F	F	0	F	F	F	0
Unfamiliar faces	Exp × ide	T	T	T	1	T	T	T	0	T	T	F	0	F	T	T	0
	Ide × exp	T	T	F	1	F	F	T	0	T	F	T	0	F	F	T	0

Note. MRI = Test of marginal response invariance; Exp = Expression; Ide = Identity; d' = Test of marginal sensitivity; c = Test of marginal response bias; T = Equality holds; F = Equality does not hold. The number in the SI column indicates the total number of failures of SI (of 16 possible).

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Table 10
Inferences Regarding PI, PS, and DS for Both Classes of Stimuli and All Four Observers (Obs.)

		Obs. 1				Obs. 2				Obs. 3				Obs. 4			
Stimulus	Comparison	PI	PS	DS	OT	PI	PS	DS	OT	PI	PS	DS	OT	PI	PS	DS	OT
Familiar faces	Ide × Exp	T	T	F	3	T	F	T	2	T	F	T	1	T	F	F	2
	Exp × Ide	T	T	T	1	T	F	T	2	T	F	F	2	T	F	F	2
Unfamiliar faces	Ide × Exp	T	T	T	1	T	T	T	1	T	T	F	1	T	T	T	2
	Exp × Ide	T	T	F	1	T	F	T	2	T	F	T	1	T	F	T	2
Number of violations		0	0	1		0	3	0		0	3	2		0	3	2	

Note. OT = Outcome type (Correspondence of the marginal analyzes and the model fits); Ide = Identity; Exp = Expression.
Outcome type (OT): 1. Identical inferences supported by marginal analyzes and model fit. 2. Model fit resolves ambiguity from marginal analyses. 3. Inferences from models and marginals differ nominally, but agree by overlapping confidence intervals on parameter estimates.

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tive of the complete ensemble of stimuli, where values of identity and expression vary from trial-to-trial, these dimensions appear to be capable of producing both perceptual, and decisional interactions.

Experiment 3

In Experiments 1 and 2, we tested for dimensional interactions between identity and expression, using Garner's speeded classification task (Experiment 1, Garner, 1974), and GRT (Experiment 2, Ashby & Townsend, 1986). These approaches were employed to assess separability and independence, which according to the parallel-route hypothesis, characterizes the encoded representations of identity and expression (Bruce & Young, 1986). The strength of these approaches is that they provide information about the statistical structure of the encoded information. They cannot, however, speak to independence in terms of the real-time characteristics of processing that encoded information. To address this type of independence, we turn to an approach developed specifically for this question, and one that has been applied to similar questions in face perception: systems factorial technology (SFT, Townsend & Nozawa, 1995; Townsend & Wenger, 2004a, 2004b; Ingvalson & Wenger, 2005; Wenger & Townsend, 2001).

SFT is a set of allied theory and methodology that has been developed to address the foundational questions of processing architecture (parallel, serial, coactive), stopping rule (self-terminating, exhaustive), effects of variations in workload (capacity, limited, unlimited, super), and the presence or absence of dependencies in rates of processing across channels of encoded information. The theoretical analyses (see in particular Townsend & Nozawa, 1995) draw on more than three decades of work on these questions (see Townsend & Wenger, 2004b, for a review) and are summarized in a set of distribution-free strong-inference predictions for a set of converging measures capable of speaking to all four of these questions. The empirical methodology follows from the theory and is known as the double-factorial paradigm (DFP). This paradigm includes a set of conditions necessary for the collection of the critical converging measures derived from the theory.

Method

Participants. Four new observers from the Pennsylvania State University, who did not participate[6] in either of the previous experiments, participated. They were compensated at the rate of $8 per hour. All of them reported normal or corrected to normal vision. Each observer completed 12 days of testing.

Materials. A key aspect of the SFT is the nested manipulation of factors relevant to inferences regarding each of the four foundational characteristics. At the outer level of this nesting are manipulations of workload—number of dimensions relevant to the response—balanced (in terms of trial frequencies) according to the decision rule used to guide responding on the basis of some or all of those dimensions. At the inner level of this nesting is a manipulation intended to selectively influence (Townsend & Nozawa, 1995; Dzhafarov, 2003; Dzhfarov, Schweickert, & Sung, 2004; Dzhafarov, 1999) the rate with which each target dimension is being processed. In this experiment, the rate of processing identity was slowed by adding sunglasses, and the rate of processing expression was slowed by adding a scarf at the mouth.[7]

The experiment was performed using the same apparatus as in the previous experiments. Stimuli consisted of the eight images used in the previous experiments. Three variations were produced for each image, resulting in a total of 24 additional images. In one image the eyes were covered with sunglasses, in another, the mouth area was hidden behind a scarf, in a third face, both changes were made (see Figure 1). Information located around the eyes is critical for recognition of identity, whereas features, such as facial muscles, curvature, and so forth, are critical for the identification of expression (see for, Ekman & Friesen, 1976; Ellison & Massaro, 1997).

In total, there were 32 gray-scale images, resulting from the crossing of two identities and two expressions, at two levels of familiarity (familiar, unfamiliar), at four levels of selective influence (Figure 1(a)) and 1(b)). Viewed at a fixed distance of 76 cm, the images subtended 1.20° of visual angle horizontally and 1.20° vertically.

Design and procedure. The two versions of the task used the same stimuli and differed only in the response required from

[6] Our intention had been to test the same observers in all three experiments. However, logistical challenges prevented these observers who participated in Experiments 1 and 2 from participating in Experiment 3.

[7] The reliability of these manipulations, and their ability to produce data that are consistent with the assumption of selective influence, was motivated by yet-to-be-published results (Wenger & Copeland, 2002) from our lab.

observers. In the first version (referred to as the AND condition), observers were instructed to give a positive response only if they perceived a specific conjunction of identity and emotion (e.g., a happy face that belonged to Keira Knightley); they were to give a negative response to all other possible combinations of identity and expression. In the second version (referred to as the or condition, observers were instructed to give a positive response if they perceived the target identity or the target expression (either a happy face, Keira Knightley, or both); all other combinations of identity and expression were to receive a negative response. Two observers participated in the AND task, and two observers performed in the or task.

The experiment was designed as a 2 (Familiarity) × 2 (Identity) × 2 (Expression) × 2 (Ambiguity) factorial design, with all factors manipulated within observers. Familiar faces and unfamiliar faces were presented in different blocks. In the OR task, each experimental block consisted of 128 trials, and every testing session contained 14 blocks. In the and task, an experimental block consisted of 144 trials, and every testing session included 12 blocks. Each observer completed 12 days of testing. Therefore, each participant performed in more than 144 experimental blocks. The 16 pictures for each level of familiarity were presented in a block according to the frequency constraints mentioned. The keyboard keys "m" and "z" were used as response keys, with the former indicating a target, and the latter signaling a nontarget. All participants were right-handed.

In all tasks, frequencies of target and nontarget trials were equated, as were the frequencies of single- and double-target trials (Mordkoff & Egeth, 1993; Mordkoff & Yantis, 1991). This practice is critical to preventing a bias in response to either of the stimuli. Half of the participants started each session with a block of familiar faces, while the other half started with a block of unfamiliar faces. Trials in all blocks were self-initiated. Trials began with a fixation point centered on the screen for 100 ms, followed by a gray-scale image of a face presented on a black background, for 70 ms. The observer then responded with one of the two specified response keys. Latency and accuracy were recorded by the computer. No feedback was provided. A 1-min break was given after each block of trials.

Assuming that selective influence holds, the conditions provided in this nested pair of factorial designs allows for inferences regarding architecture and stopping rule, independence (in rate), and capacity to be made (as established formally in Townsend & Nozawa, 1995; Townsend & Wenger, 2004b). Specifically, the four cells of the design where both dimensions of the test stimulus are in the target (positive response) state can be used to draw inferences regarding process architecture and stopping rule. These conditions are those in which the level of work (number of dimensions in the target state) is held constant while rate of processing for each dimension is varied. The three cells in which the rate of processing can be assumed to be at its highest across variations in level of work can be used to draw inferences about capacity and independence.

Results

We consider first the data with respect to the assumption of selective influence, then report on the measures that speak to the questions of architecture and stopping rule. Finally, we present the data that are relevant to inferences regarding capacity and independence. Before analysis, RTs shorter than 150 ms or longer than 2000 ms were excluded (a total of 8% of all trials across all observers) and all RT analyses were based on correct responses.

To test for consistency of the data with the assumption of selective influence, we performed a 2 × 2 ANOVA with identity (ambiguous, unambiguous) and expression (ambiguous, unambiguous) as factors. We did this separately for each observer and for the familiar and unfamiliar stimuli. Tables 11 and 12 present the results of these analyses. The occurrence of reliable main effects for each of the factors is taken as evidence that the patterning of the data are consistent with the assumption of selective influence. As can be seen in Tables 11 and 12, these main effects were reliable in all cases except for one (observer 2, unfamiliar faces), suggesting (but not guaranteeing) that selective influence of the experimental manipulations was in force.

Architecture and Stopping rule. The RT measures that are used to draw inferences about the four characteristics of processing are framed at two levels: at the level of the mean, and at the level of the distribution. Let the subscripts *A* and *U* indicate the level for each dimension (identity, expression), with *A* to indicate that the dimension was ambiguous (slow, by adding either the sunglasses or the scarf), and *U* to indicate that the dimension was unambiguous (fast). Let the order of the subscripts be state of identity first, state of expression second. Thus, $\overline{RT}_{AU}$ will refer to the mean RT when identity is ambiguous and expression is unambiguous.

The first pair of RT measures are interaction contrasts at the level of the mean and the distribution for the 2 × 2 factorial involving the ambiguity manipulation on each of the two dimensions. These interaction contrasts are identical to those used in analyses of variance for two-way designs (Maxwell & Delaney,

Table 11
Experiment 3: Selective Influence Is Examined With ANOVA on Mean RT, for Each of the Observers (O), for Familiar Faces

O	Effect	*df*	*MSE*	*F*	O	Effect	*df*	*MSE*	*F*
1	E	1	1103708.41	47.47‡	2	E	1	118259.76	7.22‡
	I	1	557870.83	23.99‡		I	1	372039.30	22.70‡
	E × I	1	239878.90	10.32‡		E × I	1	405170.33	12.76†
3	E	1	13559070.79	178.90‡	4	E	1	894075.34	31.67‡
	I	1	808421.28	10.67‡		I	1	829149.81	29.37‡
	E × I	1	157395.77	4.93*		E × I	1	139195.45	4.76*

Note. The factors are identity (ambiguous, unambiguous) and expression (ambiguous, unambiguous). E = expression; I = Identity.
* $p < .05$. † $p < .01$. ‡ $p < .001$.

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Table 12
Experiment 3: Selective Influence Is Examined With ANOVA on Mean RT, for Each of the Observers (O), for Unfamiliar Faces

O	Effect	*df*	*MSE*	*F*	O	Effect	*df*	*MSE*	*F*
1	E	1	235367.79	32.17‡	2	E	1	75902.82	5.47*
	I	1	232532.69	31.79‡		I	1	33457.08	2.41
	E × I	1	35331.01	4.83*		E × I	1	59580.72	2.51
3	E	1	5200991.02	53.77‡	4	E	1	125279.133	5.02*
	I	1	2644094	27.34‡		I	1	2565308.12	50.00‡
	E × I	1	272885.71	4.07*		E × I	1	2682092.97	5.89*

Note. The factors are identity (ambiguous, unambiguous) and expression (ambiguous, unambiguous). E = expression; I = Identity.
* $p < .05$. † $p < .01$. ‡ $p < .001$.

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2004), as well as to the interactions that were central to the approach to architecture developed by Sternberg (Sternberg, 1966, 1969a, 1969b, 1975; Roberts & Sternberg, 1993). The interaction contrast at the level of mean RT (MIC) is defined as

$$MIC = \overline{RT}_{AA} - \overline{RT}_{AU} - \overline{RT}_{UA} + \overline{RT}_{UU} \quad (1)$$

Table 14 and 15 present the RT means and error rates for each condition along with the mean interaction contrast for RTs (e.g., MIC). MIC can be either zero, positive, or negative, and combined with the pattern of survivor interaction contrast (e.g., SIC), these values guide inferences on architecture and stopping rule (Townsend & Nozawa, 1995).

The corresponding distributional contrast is framed at the level of the survivor function of the RT distribution,

$$\begin{aligned} S(t) &= P(T > t) \\ &= 1 - P(T \leq t) \\ &= 1 - F(t) \end{aligned}$$

where $F(t)$ is the cumulative distribution function. This survivor function interaction contrast (SIC) is defined as:

$$SIC = S_{AA}(t) - S_{UA}(t) - S_{AU}(t) + S_{UU}(t). \quad (2)$$

Townsend and Nozawa (1995) show, at an extremely general level (i.e., absent specific distributional assumptions, and with explicit consideration of base times), how these two interaction contrasts, when used together, allow for strong and unique inferences regarding processing architecture and stopping rule. The logic of this work is presented in Table 13. Table 13 is a truth table that guides inference on the possible architecture and stopping rule given the Mean and Survival interaction contrasts. A full exposition and formal support can be found in Townsend and Nozawa (1995). We plot the four survivor functions, and their corresponding survivor interaction contrast, for each observer and familiarity level in Figure 5. We performed Kolmogorov–Smirnov tests on those distributions to verify that they are ordered such that $S_{AA} < S_{AU}$, $S_{UA} < S_{UU}$. All survival functions were ordered according to this test excepting those for observer 2, unfamiliar faces.

Before proceeding with the remaining analyses, it is worth mentioning that the task at hand constrains inference regarding possible stopping rules. Recall that observers 1 and 2 performed in an OR task, whereas observers 3 and 4 performed in an AND task. Therefore, observers 1 and 2 could have used either exhaustive or self-terminating rule, whereas observers 3 and 4 must have used an exhaustive rule. This being acknowledged, we are now ready to draw inferences regarding architecture and stopping rule by combining evidence from the MIC and the SIC. We do so for each observer, for each type of familiarity, separately. Note that observer's two unfamiliar faces results were precluded from this process, because selective influence did not hold in their mean RTs.

Observer 1 showed a positive MIC (49 ms, $p < .001$, 18 ms, $p < .05$) and a positive SIC for both familiar and unfamiliar faces, indicating an independent parallel self-terminating architecture and stopping rule. Observer 2 exhibited a positive MIC (23 ms, $p < .001$) and a positive SIC for familiar faces, pointing to the same conclusion. According to these results, when processed in an OR task, identity and expression do not interact whether faces are familiar or unfamiliar. Processing finishes when information about either dimensions reaches a response criterion. These results are in accordance with hypotheses 3(a) and 3(b), which predicted parallel architecture with a self-terminating or exhaustive stopping rule.

Observer 3 exhibited positive MIC (13 ms, $p < .05$), and negative to positive SIC for familiar faces, indicating a coactive architecture. The same observer showed a negative MIC (−53 ms, $p < .05$) and a negative SIC for unfamiliar faces. This combination

Table 13
Summary of Mean (MIC) and Survivor Function Interaction Contrasts (SIC) for Alternative Architectures and Stopping Rules

Architecture	Stopping rule	MIC	SIC(t)
Serial	Self-terminating	0, Additive	Zero (flat)
Serial	Exhaustive	0, Additive	Negative to Positive
Parallel	Self-terminating	>0, Over-additive	Positive
Parallel	Exhaustive	<0, Under-additive	Negative
Coactive	n/a	>0, Over-additive	Negative to Positive

Table 14
Summary of Means for the Selective Influence Manipulation, With Familiar (Top) and Unfamiliar (Bottom) Faces

	Observer			
Condition	O1	O2	O3	O4
I0	367	393	603	325
E0	380	415	625	344
I1	396	400	751	353
E1	458	445	786	402
MIC	49	23	13	30
I0	310	330	651	472
E0	324	335	760	557
I1	324	340	793	494
E1	356	354	849	607
MIC	18	9	−53	28

Note. The letters "I" and "E" refer to identity and expression, respectively. The digits "0" and "1" refer to unambiguous and ambiguous conditions, respectively. MIC is the mean interaction contrast resulting from the means.

points to a parallel, exhaustive architecture and stopping rule. Observer 4 showed a positive MIC (30 ms, $p < .05$) and a negative to positive SIC, for familiar faces, implying a coactivative architecture. This observer exhibit a similar pattern for familiar faces with a positive MIC (28 ms, $p < .05$) and negative to positive SIC. In an OR task, identity and expression are processed in parallel. However, in an AND task, these dimensions seem to interact. These results are not fully consistent with hypotheses 3(a) and 3(b).

Capacity and independence. The distributional measure that allows for inferences regarding capacity and independence is the capacity coefficient $C(t)$. The initial version of $C(t)$ (Townsend & Nozawa, 1995) was developed for tasks using the or response rule. Later work (Townsend & Wenger, 2004b) extended the conception to the and response rule. In both cases, the index is constructed as a benchmark, with the basis of comparison being parallel independent unlimited capacity processing. For the or response rule, the capacity coefficient is defined as

$$C_O(t) = \frac{H_{EI}(t)}{H_E(t) + H_I(t)} \tag{3}$$

where

$$H(t) = \int_0^\infty h(t)dt = -\ln[S(t)]$$

is the integrated hazard function, a measure of the cumulative level of work accomplished by time t (Townsend & Ashby, 1978; Townsend & Nozawa, 1995; Wenger & Townsend, 2000, 2006), which is equal to the negative log of the survivor function. Here the subscript E indicates the condition in which only expression is processed, I stands for the condition in which only identity is processed, and EI refers to the condition in which both identity and expression are processed. For the and response rule, the capacity coefficient is defined as

$$C_A(t) = \frac{K_E(t) + K_I(t)}{K_{EI}(t)} \tag{4}$$

where

$$K(t) = -\ln[F(t)]$$

(Townsend & Wenger, 2004b). For both versions of the capacity coefficient, a value of 1 corresponds to unlimited capacity processing and is the value that is obtained under the assumption of independence in rate. Values less than 1 indicate limited capacity processing, and values greater than 1 indicate supercapacity processing. This latter outcome is important as, if the interaction contrasts support coactive processing rather parallel processing, (Mordkoff & Yantis, 1991; Mordkoff & Egeth, 1993; Townsend & Nozawa, 1995), then this is a strong indication of a violation of independence in rate (see, e.g., Ingvalson & Wenger, 2005).

Both the OR and AND capacity coefficients are related to two well-known inequalities in the RT literature (Townsend & Nozawa, 1995; Townsend & Wenger, 2004b). In the case of the OR coefficient, the Miller (Miller, 1982)

$$S_{EI}(t) - S_E(t) - S_I(t) + 1 \geq 0 \tag{5}$$

and Grice (Grice, Canham, & Gwynne, 1984)

$$\min[S_E(t), S_I(t)] - S_{EI}(t) \geq 0 \tag{6}$$

inequalities provide extreme upper (supercapacity) and lower (limited capacity) bounds on the interpretation of $C_o(t)$. In the case of the AND coefficient, the corresponding bounds are provided by Colonius and Vorberg (Colonius & Vorberg, 1994)

$$F_E(t) + F_I(t) - 1 \leq F_{EI}(t) \leq \min[F_E(t), F_I(t)] \tag{7}$$

(Townsend & Wenger, 2004a, 2004b).

Figure 6 presents the capacity coefficients and the lower and upper boundaries (the Miller and Grice inequalities for the OR task, the other for the AND task). First, consider the four upper

Table 15
Summary of the Probability for Correct (P(Correct)) in the Selective Influence Manipulation, With Familiar (Top) and Unfamiliar (Bottom) Faces

	Observer			
Condition	O1	O2	O3	O4
I0	0.97	0.97	0.98	0.98
E0	0.91	0.93	0.97	0.96
I1	0.95	0.95	0.96	0.97
E1	0.88	0.95	0.93	0.94
I0	0.97	0.93	0.98	0.95
E0	0.95	0.92	0.97	0.88
I1	0.97	0.95	0.96	0.97
E1	0.94	0.92	0.91	0.92

Note. The letters "I" and "E" refer to identity and expression, respectively. The digits "0" and "1" refer to unambiguous and ambiguous conditions, respectively. MIC is the mean interaction contrast resulting from the means.

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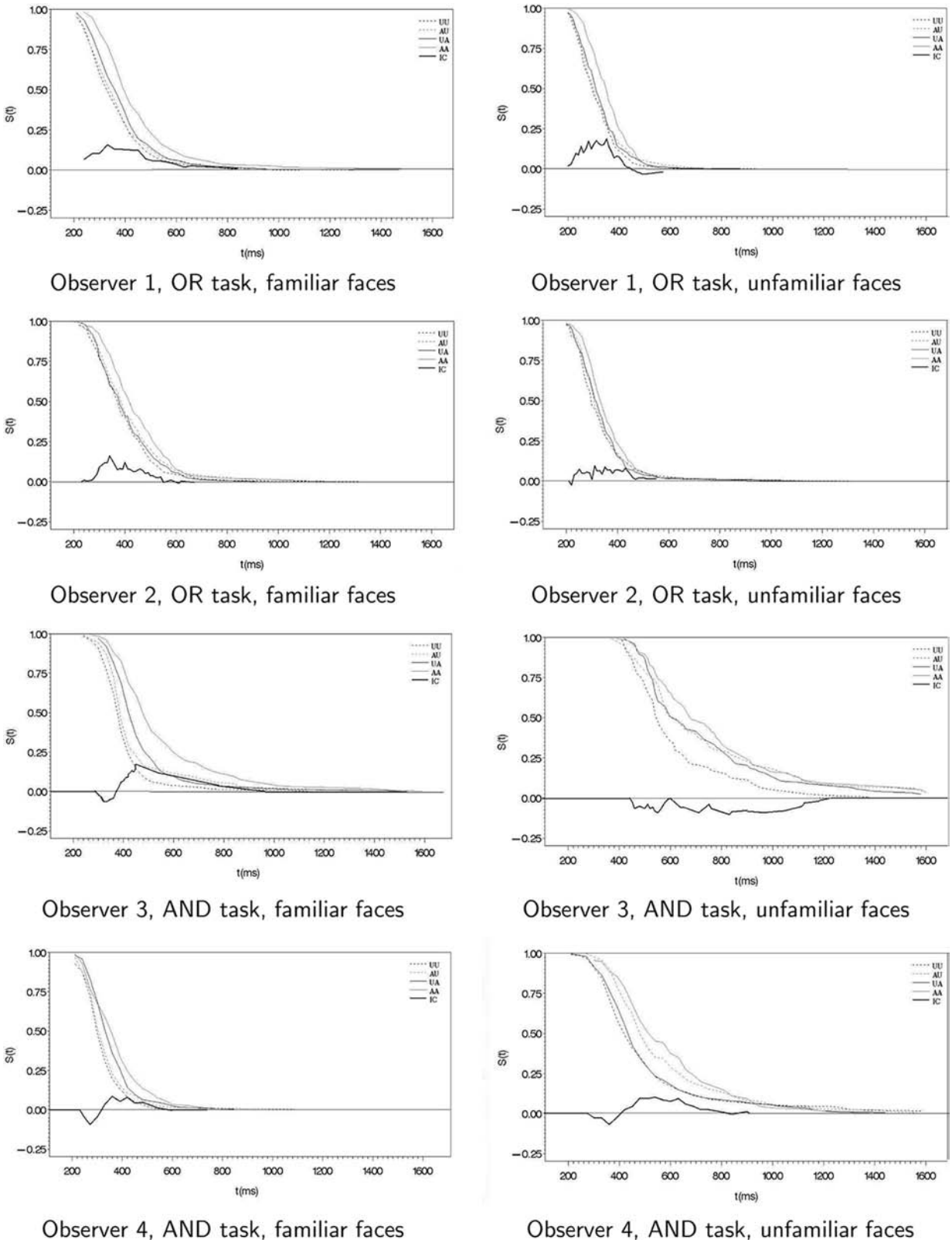


Figure 5. Experiment 3: A plot of the survival interaction and survival functions where $S(t)$ is the survivor function for task-completion times, against t (time) for four conditions of selective influence. U = Unambiguous; A = Ambiguous. The order of these letters is such that the first stands for identity and the second for expression.

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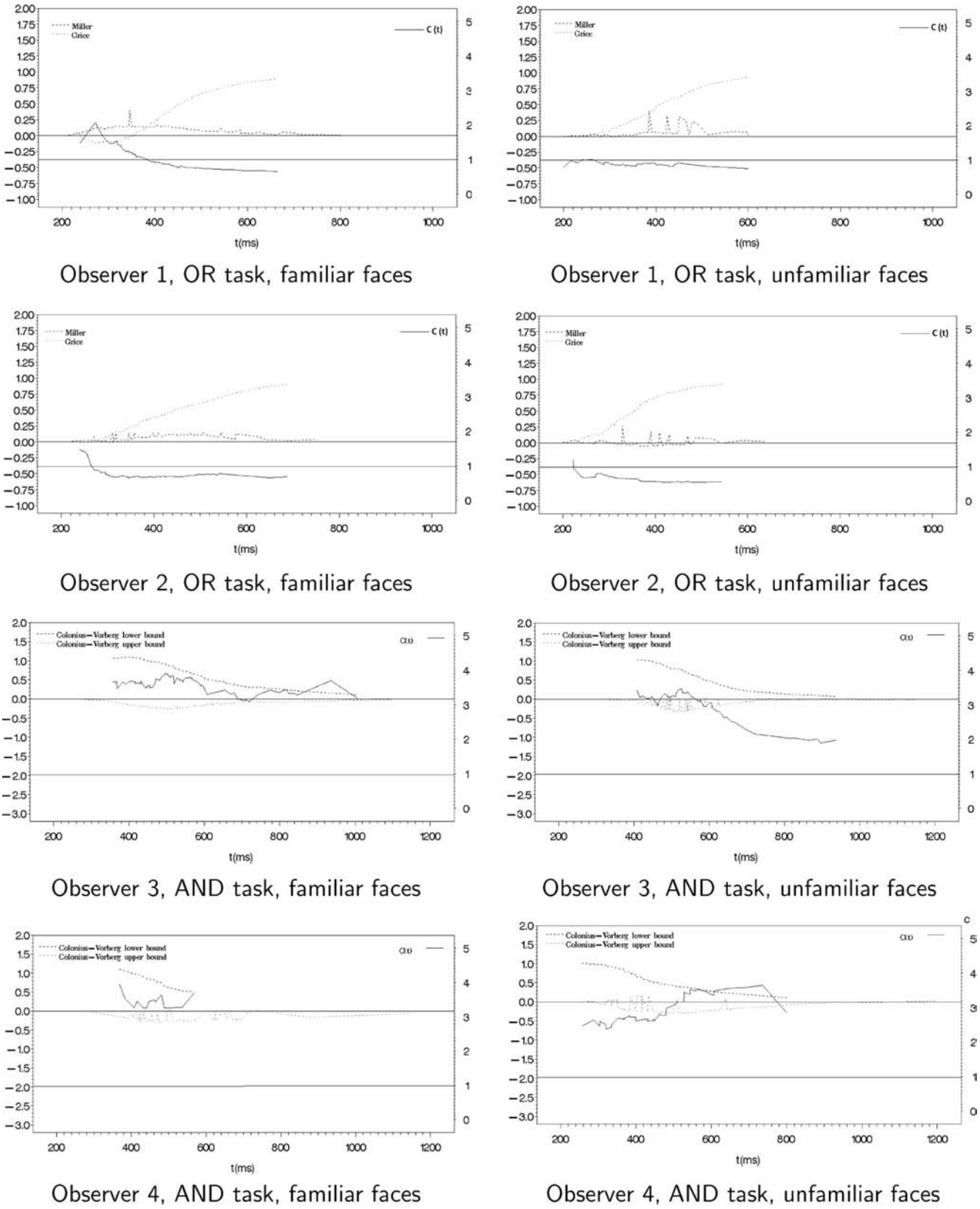


Figure 6. Experiment 3: A plot of the capacity coefficients and the lower and upper boundaries (the Miller and Grice inequalities for the OR task, and the Colonius/Vorberg inequalities for the AND task).

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plots describing observers 1 and 2 data with respect to familiar and unfamiliar faces. As can be noted, the capacity coefficient $C_O(t)$ in all cases is either equal to or smaller than 1. This reflects unlimited to limited capacity, a conclusion that receives further support from the fact that neither the Grice or the Miller inequalities were violated. These results are consistent with hypotheses 3(c) and 3(d), which predicted unlimited capacity and independence in rate.

Next, consider the four lower plots describing observers 3 and 4 data with respect to familiar and unfamiliar faces. The capacity coefficient here, $C_A(t)$, is virtually larger than 1. This result implies supercapacity processing, an outcome that is further supported by the observation of violations of both the upper and lower bounds of the Colonius/Vorberg inequalities. The outcomes for observers 3 and 4 do not support hypotheses 3(c) and 3(d), which predicted unlimited capacity and independence in rate.

Taken together, these results suggest that when processed in an OR task, identity and expression are independent in rate, and producing unlimited to limited capacity. However, when processed together (e.g., in an AND task) these pair of dimensions have positive or negative dependencies, which lead to supercapacity. Note that we also found coactivation architecture with familiar faces in this task. This finding may suggest that the dependencies are the result of the pooling of evidence into a single decision mechanism that results in an increase in capacity.

Discussion

The goal of Experiment 3 was to test hypotheses 3(a)–3(d), which refer to architecture, stopping rule, capacity, and independence. The combined evidence showed that hypotheses 3(a)–3(d) were supported in the *OR* task, but refuted in the *AND* task. This suggests that identity and expression are independent to some extent, depending on whether observers are asked to integrate information from the two dimensions, or whether they are required to respond to either of the dimensions. The parallel-route hypothesis cannot accommodate the former results.

General Discussion

Taken as a whole, the evidence adduced in the current study for the parallel-route hypothesis (Bruce & Young, 1986) is weak to medium. Given the fact that a variety of measures of independence have been used in this study, we submit that independence in both the way identity and expression are represented, and in the rate at which these dimensions are processed, is violated.

Implications for Neurobiological and Cognitive Face Recognition Models

Part of the theoretical motivation for studying the parallel-route hypothesis was initially sparked by selective brain damage studies (Calder & Young, 2005). In particular, reports on patients who showed impaired facial identity recognition and intact facial expression recognition, along with other patients who exhibited a reverse dissociation (Etcoff, 1984; Parry et al., 1991; Tranel, Damasio, & Damasio, 1988). This apparent double-dissociation (in conjunction with other lines of evidence) has lead researchers to posit functional (Bruce & Young, 1986) and neurobiological (Haxby et al., 2000) separation in the processing of identity and expression. Recently, Calder and Young (2005) have persuasively criticized the logic and validity of those brain damage studies and consequently cast doubt on the parallel-route hypothesis: "recognition of facial identity but not facial expression can occur in neuropsychological conditions other than selective damage (or disrupted access) to the perceptual representation of human face (prosopagnosia), and these causes of the dissociation do not necessarily support separate visuoperceptual codes for facial identity and expression" (p. 642). If the evidence from brain-damage patients is ambiguous, then additional well-controlled behavioral studies with healthy patients should be performed, as has been done in the current effort.

In our view, the difficulty in interpreting the parallel-route hypothesis is rooted in a paradox of selective attention (Melara & Algom, 2003). On the one hand, recognition of facial identity is critical for survival and therefore must not be affected by facial expression (Haxby et al., 2000). On the other hand, recognition of facial identity and expression rely on overlapping perceptual spatial features, and therefore information about one dimension is valuable when recognizing the other dimension. Our approach resolves the tension between conflicting requirements for selective attention and the need for integration of information from the two dimensions (Melara & Algom, 2003) by assuming the existence of various levels of (in)dependence. Hence, good attentional selectivity (i.e., lack of interaction between identity and expression) occurs with respect to some forms of independence, but not with others. This can explain the mixture of conflicting results, some generally supportive of the parallel-route hypothesis, others much less so.

To demonstrate the merits of our approach consider the findings from Experiment 1 and Experiment 2. We found preservations of perceptual independence (PI), a form of within-stimulus independence, in tandem with violations of perceptual separability (PS) and Garner interferences, both forms of cross-stimuli dependency. The implications of these outcomes for everyday perception of faces is that at the single face level, facial identity is not affected by the variant properties of expression. However, at the stimuli ensemble level, identity and expression interact.

This implies the biological plausibility of neurons that encode either identity or expression, along with other neurons that encode both dimensions. Such a possibility is espoused by the PCA approach (Calder & Young, 2005; Calder et al., 2001).

The application of the Garner paradigm to faces typically uses a small sample of stimuli (but see, Ganel & Goshen-Gottstein, 2002), which might allow participants to develop strategies associated with the properties of the specific images rather than of the larger classes of which these are exemplars. We believe that this does not undermine the results of the current effort but acknowledge the limitation that the current work shares with previous published papers using the Garner paradigm. A recent account (Burton, Jenkins, & Schweinberger, 2011; Jenkins, White, Montfort, & Burton, 2011) suggests that image variability is an important variable in face recognition so that face units (FRUs) should not be regarded "abstract" representation of a face. Representations for face identification need to explicitly incorporate idiosyncratic within-person variability. Future work should address the issue of image variability and its impact on the question of independence of facial dimensions.

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Implications for Theories of Perceptual Independence

Perceptual independence is an important construct in cognitive psychology. One example is the distinction made between separable and integral dimensions (Garner, 1974; Nosofsky, 1986; Shepard, 1964). The work of Garner and Morton (Garner & Morton, 1969) has been a cornerstone in shaping many of the related concepts. Garner and Morton intuited that various measures and models of statistical independence should converge (Garner et al., 1956). Later, attempts at developing coherent and quantitative theories of perceptual independence have been made (Ashby & Townsend, 1986; Ashby, 1989; Nosofsky & Palmeri, 1997). The goal of unification and relating different theories of independence has also been pursued (Ashby & Perrin, 1988; Ashby & Lee, 1991). Ashby and Maddox have proposed a model that related GRT to Garner's speeded classification task (Ashby & Maddox, 1994; Maddox, 1992). An alternative model has been proposed by Nosofsky and Palmeri (1997) who applied the EBRW—a random-walk extension of the GCM model (Nosofsky, 1986)—to RT data in a Garner task. Recently, Fific, Nosofsky, and Townsend (2008) have validated the SFT with separable and integral dimensions. These authors have shown that with separable dimensions observers exhibit a serial or parallel architecture with an exhaustive stopping rule, whereas with integral dimensions they demonstrate coactivation. In yet another recent study, Fific, Little, and Nosofsky (2010) proposed a rule-based model that interrelates GRT to both RTs and accuracies. In accordance with this work, our results showed high level of consistency across various definitions of independence. Future work should develop models that relate the various definitions of perceptual independence into a coherent and unified conceptual framework.

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Appendix

Methods of Fitting and Evaluating the Bivariate Gaussian Models

The model-fitting and evaluation procedure used here was first proposed by Ashby (1992), and was further developed by Thomas (2001b) and Wenger (Copeland & Wenger, 2006; Cornes et al., 2011). The procedure is performed by defining a hierarchy of possible models, beginning with a null model in which PI, PS, and DS hold. The hierarchy of models is then extended by systematically increasing the number of free parameters. Varieties of potential violations are captured by these models. Their development is guided by the results of the marginal analyses. In that sense, the fitting processes is not a "complete guess" as in other model fittings but is constrained by additional evidence coming from the same set of confusion data (see for, Ashby & Lee, 1991). Each model in the hierarchy is nested within another model, posited higher in the branch.

To perform the procedure efficiently we adopted the following assumptions (see also, Copeland & Wenger, 2006; Cornes et al., 2011; Thomas, 2001a): (a) equal variance for all distributions (equal to 1), (b) the location of one of the distributions (e.g., person A, happy) was fixed to have marginal means of (0,0), (c) decision bounds were assumed linear continuous or piecewise (depending whether the model assumed DS or not). The simplest model thus contains four free parameters with the following mean vectors and covariance matrices:

$$\mu_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mu_2 = \begin{bmatrix} 0 \\ \mu_{2,2} \end{bmatrix}, \mu_3 = \begin{bmatrix} \mu_{3,1} \\ 0 \end{bmatrix}, \mu_4 = \begin{bmatrix} \mu_{3,1} \\ \mu_{2,2} \end{bmatrix},$$

where the $\mu_{i,j}$s are parameters to be estimated. Note here the equalities used to represent PS holding. Each of the four (i = 1. . .4) variance/covariance matrices for this model would be set to:

$$\Sigma_i = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

and the two decision bounds would be specified as two parameters to be estimated, γ_1 and γ_2. Thus this model requires four free parameters.

We fit the models to the empirical identification confusion matrices, separately for each familiarity level, for each observer (a total of eight confusions matrices), using methods developed by Macho (2007). Decisions about the best fitting model were based on the log likelihood for the model along with measures of fit (Akaike Information Criterion, AIC, Akaike, 1983; Thomas, 2001a), adjusted for the number of free parameters. The χ^2 statistic was used to compare models in hierarchical relationship, with degrees of freedom equal to the difference in the number of free parameters. The AIC value was used to decide between models not in a hierarchical relationship. Lower value of AIC indicated superior model.

Received June 12, 2011
Revision received January 24, 2012
Accepted January 24, 2012 ■

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